Data Engineering

Creating and Managing Virtual Environments in VS Code

Course: Data Engineering Foundations & Pipeline Architecture

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1 Method 1: VS Code Command Palette (Recommended)

Visual Studio Code provides automated tooling that initializes the virtual environment, configures internal workspace settings, and associates the Python interpreter in a single operation.

1. **Open the Project Workspace:** Open your project's root directory via **File** → **Open Folder...**.
2. **Launch the Command Palette:** Press **Ctrl + Shift + P** (Windows/Linux) or **Cmd + Shift + P** (macOS).
3. **Trigger Creation Command:** Type and select: `Python: Create Environment...`
4. **Select Environment Type:** Choose **Venv**.
5. **Choose the Base Python Interpreter:** Select the Python executable installed on your system.

2 Method 2: Using the Integrated Terminal

Step 1: Open Terminal

Open the integrated terminal (**Ctrl + '** or **Terminal** → **New Terminal**). Ensure the current working directory is the root of your project.

Step 2: Generate the Environment

```
# Windows
python -m venv .venv

# macOS / Linux
python3 -m venv .venv
```

Step 3: Activate the Environment

OS	Shell	Activation Command
Windows	PowerShell	`.\.venv\Scripts\Activate.ps1`
Windows	CMD	`.\.venv\Scripts\activate.bat`
macOS / Linux	Bash / Zsh	`source .venv/bin/activate`

PowerShell Note: If Windows blocks script execution, run: `Set-ExecutionPolicy -ExecutionPolicy RemoteSigned -Scope Process`

3 Configuring the Interpreter in VS Code

To guarantee that IntelliSense and code navigation point to your isolated packages:

1. Press **Ctrl + Shift + P** (or **Cmd + Shift + P**).
2. Type and select: **Python: Select Interpreter**.
3. Choose the interpreter path located within your project directory (e.g., `.\.venv\Scripts\python.exe` or `./.venv/bin/python`).